Networking Cheatsheet

By Sundeep kumar ...

Basic

1. What is Computer Networking
2. Type of transmission Media
3. Computer Network Devices
4. Unicast, BroadCast and Multicast
5. LAN vs MAN vs WAN
6. what is Web ? ---- Difference between Web vs Internet
7. Networking Topology
    - Mesh
    - Star
    - Bus
    - Ring
    - Tree
8. Basic terms:
    - Client
    - Server
    - Peer
    - Host
    - BandWidth
    - Jitter
    - Packet
    - Frame
    - Local Host
    - Bit rate
    - noise
    - attenuation
    - distortion

OSI Model

1. What is OSI Model
2. How a PACKET travels ? [V.V.Very Important
3. Different Layers ----- PDNTSPA -

    1. Physical
    2. Data Link
    3. Session
    4. Presentation
        - What is Ethernet ?
        - Extranet vs Intranet vs Internet
        - Virtual ckt vs Datagram Network
        - Switch & its types
        - What is Virtual LAN ?
        - Basics of Wifi
    5. Network
        - IP Addressing ??
        - IPv4 vs IPv6
        - Supernet
        - Network Address Translation, NAT
        - WPA vs WPS
        - LiFl vs WiFi
        - What is ACL [Access-list] ?
        - AAA in networking [important]
        - SONET
        - Types of Routing

            Static

            Default

            Dynamic

6. Transpart
    - TCP Connection Setup ------- 3-way Handshake
    - UCP
    - Peer2Peer Sharing
    - Congestion in Computer Networks
    - Error Control

7. Application
    - Protocals in Application
    - DNS
    - What does DNS use UDP and net TCP ?
    - Address Resolution in DNS (Domain Name Server)
    - DNS Spoofing or DNS Cache poisoning
    - Types of DNS Attcaks
    - Difference between http:// and https://
    - How DHCP server dynamically assigns IP address to a host

Misc & System Design

1. Difference between http:// and https://
2. What is API Gateway ?
3. SSL/TLS
4. Reverse Proxy ?
5. Load Balancer
6. ARP, Address Resolution Protocol
7. Horizontal vs Vertical Scaling

8. Caching & How does a website is cached
9. what is VIP in computer networks
10. REST API vs HTTP API
11. what is a Container in computer networks
12. Containerisation vs Scalability
13. Latency vs Throughput
14. 2G vs 3G vs 4G vs 5G
15. How VPN works ?
16. Gateway vs Router
17. NIC and MAC Address
18. Public vs Private IP Address
19. What is Multiplexing
20. Modem vs Router
21. How Bluetooth works
22. How Hotspot works
23. How Email works
24. How file transfer works
25. How ATM works
26. Common Networking commands:
    - ping
    - netstat
    - tracert
    - ipconfig
    - nslookup
    - route
    - pathping
    - netDiag
    - hostname
    - arp

Security

1. What is Firewall ?
2. Type of Firewall
3. Possible attacks on Firewall
4. Basic Network Attacks
5. Denial of Service and Prevention
6. Digital Signature & Certificates
7. Active & Passive attacks in Infosec
8. Types of Email Attacks
9. Term to know:
    - Cryptography
    - Symmetrics
    - Assymetric Key
    - Hashing
    - Cryptanalysis
    - AES/DES/RSA/MD5 hashing